

G. PUSHPA SATYA SAI REDDY

GenAI Engineer | AI Agent Systems

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SUMMARY

GenAI & AI Engineer who builds autonomous AI agents and production-grade AI systems that replace manual workflows. Experienced in designing end-to-end GenAI pipelines, multi-agent architectures, and scalable SaaS platforms on cloud infrastructure, with real-world deployments serving users at high reliability.

SKILLS

Languages:	Python, TypeScript, JavaScript, SQL
AI / GenAI:	Prompt Engineering, RAG, AI Agents, Multimodal AI, Google Gemini, Azure OpenAI (GPT-4o), Embeddings & Vector Search
Backend:	FastAPI, Node.js, Express, Next.js
Frontend:	React, Next.js, TypeScript, Tailwind CSS, WebSocket / WebRTC
Databases:	PostgreSQL, MongoDB, Supabase
Cloud & Deployment:	Render, Docker, GitHub CI/CD

CORE STRENGTHS

- ▶ End-to-end AI system design (data → model → deployment)
- ▶ Retrieval-Augmented Generation (RAG) architecture
- ▶ Multi-agent & agentic workflow design
- ▶ Prompt engineering & LLM evaluation
- ▶ Multimodal AI integration (text, image, voice, video)
- ▶ API integration & scalable backend design

PROJECTS

CodeHub — Real-Time Collaborative Code Editor (*React, TypeScript, Node.js, FastAPI, MongoDB, WebSocket, WebRTC, Docker*) [Live ↗](#)

- Built a full-stack browser IDE for real-time multiplayer code editing with live cursors using Yjs CRDTs over WebSockets, plus sandboxed execution of 10+ languages in isolated Docker containers.
- Integrated an AI pair-programmer (Claude/Groq) for inline completions, refactors, and error explanations, alongside a GitHub workflow (push, multi-file PRs, repo import) and in-editor WebRTC video calls with screen sharing.
- Designed a hackathon platform with team workspaces and a hybrid AI/host/community judging engine; deployed on Render + MongoDB Atlas with JWT auth and OAuth, plus a second Python/FastAPI backend sharing the same database and auth.

Habitly — AI-Powered Health & Habit Companion (*Next.js 14, Supabase, Tailwind, Google Gemini, Three.js, PWA*) [Live ↗](#)

- Built a full-stack health & habit PWA (Next.js 14 + Supabase) unifying habits, medications, weight, calories, and vitals for chronic-condition self-care; deployed on Render.
- Trained per-habit linear-regression models in plain JavaScript on a 4,000-profile synthetic dataset (R^2 up to 0.96) to generate personalized targets fully in-browser.
- Integrated Google Gemini (multimodal) to turn uploaded medical reports into structured wellness plans, and built timezone-aware Web Push reminders via a service worker + cron.

AI-Powered IPTV Chatbot with RAG System (*Azure OpenAI GPT-4o, Express.js*)

- Built a RAG customer-support system (Azure OpenAI GPT-4o + text-embedding-ada-002) with a text-chunking pipeline and cosine-similarity search over 1,536-dimensional vectors.

- Added dual-threshold anti-hallucination filtering and an Express.js backend (8 endpoints) with admin analytics — achieving 87% accuracy, sub-second retrieval, and zero hallucinations.

EDUCATION

Parul University, India	B.Tech, Artificial Intelligence • Expected Nov 2027
Pragathi College, Tanuku	Intermediate • 2023
Montessori School, Tanuku	SSC • 2021